

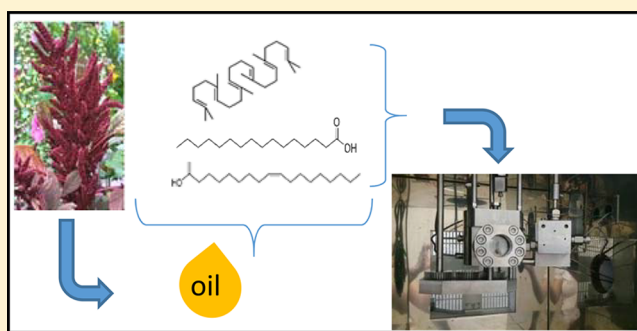
Solubility of Squalene and Fatty Acids in Carbon Dioxide at Supercritical Conditions: Binary and Ternary Systems

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ABSTRACT: Experimental solubility determinations of binary and ternary systems that involve fatty acids (palmitic acid and oleic acid) in supercritical carbon dioxide (sc-CO₂) were performed using an experimental system based on the static synthetic method with online sampling. The solubility data of squalene + sc-CO₂ and palmitic acid + sc-CO₂ were obtained and compared with data from the international literature resulting in the validation of the experimental method. Experimental solubility measurements of squalene + palmitic acid + sc-CO₂ and squalene + oleic acid + sc-CO₂ were studied at pressures from 9.00 to 30.38 MPa and temperatures between 313.30 and 333.48 K. The standard uncertainties for temperature and pressure were evaluated to be 0.02 K and 0.02 MPa, respectively. The relative combined expanded uncertainty ($k = 2$) for the composition was estimated to be $U_r(y) = 0.051$. Chrastil, del-Valle and Aguilera, Bartle et al., and Kumar and Johnston models were used to correlate the experimental solubility data. The modeling results validated the self-consistency of experimental data in the entire range of measurement.



1. INTRODUCTION

Amaranth, an ancient crop from South American cultures, was extensively cultivated because of its alimentary and religious significance.¹ Amaranth seeds have been the target of researches of composition and application of novel techniques on grain, which may cause wider cultivation and utilization of valuable components on human related products.² Amaranth oil has been reported to be constituted mainly by fatty acids such as oleic acid and palmitic acid, and in lower concentration, squalene.³

Oleic acid (OA) is a monounsaturated fatty acid (C18:1) present in vegetal oils,⁴ widely known by its health benefits and recommended to prevent diseases as cancer but its mechanism is not completely known;^{5–7} it is also a raw material for bio products,⁸ and one common technique applied to extract oleic acid is fractional distillation.⁹

Palmitic acid (PA) is a saturated fatty acid (C16:0) and limited in a daily diet to decrease possible health risks.¹⁰ It is very common in unprocessed fats.¹¹ PA is less disposed to oxidation than unsaturated compounds and crystallizes at a relatively high temperature.¹²

Squalene is a long hydrocarbonated chain (C₃₀H₅₀) found in vegetables and animal tissues due its role as an intermediate of phytoosterols and cholesterol.¹³ Several investigations have reported the biological importance of squalene as antioxidant,^{14–16} chemopreventive,¹⁷ and other biological activ-

ities.^{18–23} In the last century, shark liver oil was considered the greatest source of squalene,²⁴ but nowadays vegetable squalene sources are used.²⁵ Among the vegetable sources with the highest squalene content are amaranth seeds.^{25,26}

Czaplicki et al.²⁶ compared the content of squalene in amaranth oils obtained by three techniques: expeller pressing, organic solvent extraction, and supercritical fluid extraction (SCFE) and found that the oil obtained as a result of supercritical extraction has the highest squalene content. Advantages of SCFE are extracts and matrix being solvent-free due to no additional solvent involved in the processes, especially for residues for food purposes after extraction.²⁷ Besides, this technique works at relatively low temperatures, which is suitable for processing thermo-labile compounds. Supercritical carbon dioxide (sc-CO₂) is commonly used at SCFE because it is an inert,^{28,29} eco-friendly,²⁸ economic, and safe compound.²⁹ Solubility determination is the basis to develop SCFE processes.³⁰

Squalene solubility in sc-CO₂ has been described previously by several authors.^{31–35} However, among these studies deviations have been found, especially at high pressures and can be attributed to differences on purity of the sample,

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methodology, and/or apparatus differences.³² Since that system is of interest in this work, experimental solubilities were determined and reported, allowing comparisons between the data obtained in this work and the literature mentioned previously. Due to differences between the data obtained in this work and that of other authors, the solubility data of the palmitic acid + sc-CO₂ system were determined and compared with the international literature.^{36–38} Also, in order to study the behavior of squalene, oleic acid, and palmitic acid at high pressures in the determination of solubility in sc-CO₂, two ternary systems involving the compounds mentioned before were performed.

Chrastil,³⁹ del-Valle and Aguilera,⁴⁰ Bartle et al.,⁴¹ and Kumar and Johnston⁴² models were used to correlate the data obtained for all systems over the entire experimental conditions, the result being that all of these correlations are able to satisfy the self-consistency test.

2. METHODOLOGY

2.1. Materials and Suppliers Information. Carbon dioxide (CAS No. 124-38-9) with purity 99.995% was purchased from Infra (México). Squalene (CAS No. 111-02-4) with purity 99.0%, palmitic acid (CAS No. 57-10-3) with purity 99.0% and oleic acid (CAS No. 112-80-1) with purity 99.9% were provided by Sigma-Aldrich. Without CO₂, all compounds were carefully degassed under vacuum prior to injection into the system. The source and purity of the samples used in this work are shown in Table 1. The physical properties of all samples used in this work are shown in Table 2.

Table 1. Sample Information

compound	supplier	purity ^a (%)	purification method	analysis method ^b
squalene	Sigma	99.0	none	GC
palmitic acid	Sigma	99.0	none	GC
oleic acid	Sigma	99.9	none	GC
carbon dioxide	INFRA	99.995	none	GC

^aProvided by supplier. ^bGas chromatography.

2.2. Equipment and Experimental Procedure. The solubility measurements for binary and ternary systems at pressures up to 30 MPa were performed in equipment based on the static-synthetic method. The experimental equipment and

procedure used in this study has been detailed described previously,⁴³ therefore only a summary is presented below. A schematic of the experimental equipment is shown in Figure 1.

For binary and ternary systems, 5 mL of pure liquid (squalene, oleic acid and palmitic acid) were placed into a high pressure cell, which has two sapphire windows coupled. The experimental system was evacuated to remove all of the impurities using a vacuum pump. The carbon dioxide was fed into the system by a syringe pump. Then, the binary or ternary system was magnetically stirred. When the temperature in the cell reached and maintained the desired value for at least 8 h, and the fluctuations of the temperature and pressure in the cell were less than ± 0.01 K and ± 0.01 MPa, respectively, for at least 30 min, the equilibrium state was considered to be established; the pressure and temperature data were recorded and the vapor composition at the equilibrium state were measured by the HPLC. At least three consecutive samples with repeatability within 2% were taken, and the average value was reported as the solubility. To estimate mole fractions, a relationship between the chromatographic peak areas and known-concentration solutions were established from (0–500 $\mu\text{g}\cdot\text{mL}^{-1}$). In this work, the composition measurements were associated with two types of experimental uncertainties:⁴⁴ (1) repeatability of the HPLC data, determined to be 0.0153 mole fraction; (2) HPLC calibration, determined to be 0.0095 mole fraction. Therefore, the relative combined expanded uncertainty ($k = 2$) for the composition was estimated to be $U_c(y) = 0.051$.

The temperature was regulated by an air bath and measured with platinum probes connected to a digital indicator previously calibrated. The pressure was monitored by a pressure transducer connected to a digital indicator previously calibrated. The standard uncertainties for temperature and pressure were estimated to be 0.02 K and 0.02 MPa, respectively.

2.3. Sample Analysis. Composition analysis was carried out in high performance liquid chromatograph (HPLC) at 208 and 205 nm. The appropriate conditions of each HPLC method are mentioned below: acetonitrile and acetone (60:40% volume) for squalene; acetonitrile, methanol, and hexane (80:12:8% volume) acidified with 0.1% volume of acetic acid for fatty acids. The HPLC method for squalene and fatty acids mixtures is by gradient and is detailed in Table 3.

Table 2. Physical Properties of Compounds Used in this Work

Name	Condensed Formula	Melting point ^a / K	Boiling point ^b / K	Molecular Weight ($\text{g}\cdot\text{mol}^{-1}$) ^b	Structure
Squalene	C ₃₀ H ₅₀	198.15	558.15	410.72	
Palmitic acid	C ₁₆ H ₃₂ O ₂	337.15	544.65	256.42	
Oleic acid	C ₁₈ H ₃₄ O ₂	277.15	467.65	282.46	

^aRef 48. ^bReference at fixed pressure 3.3×10^{-5} , 13.3×10^{-5} , 1.6×10^{-6} MPa respectively).⁴⁹

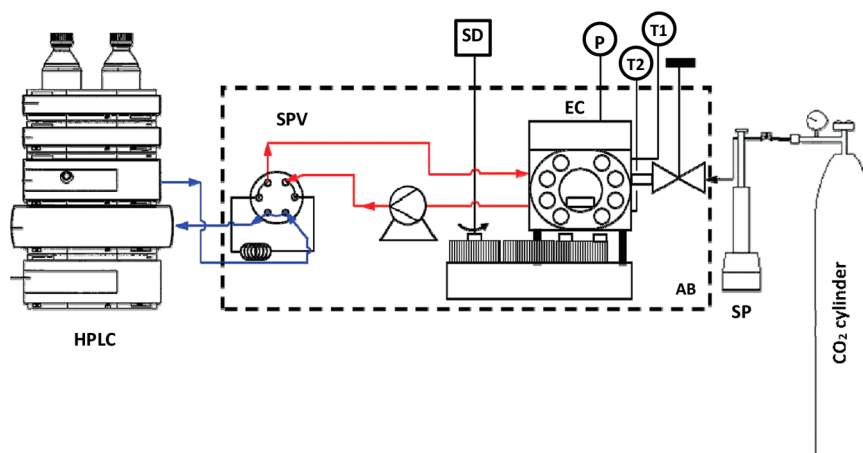


Figure 1. Schematic diagram of the experimental system: (AB) air bath, (EC) equilibrium cell, (P) pressure transducer, (SD) stirring device, (SP) syringe pump, (SPV) six-port valve, (T1 and T2) platinum resistance thermometers, (HPLC) high-pressure liquid chromatograph.

Table 3. HPLC Method (Gradient) Used To Determinate Compositions of Squalene and Fatty Acids Mixtures

time (min)	acetonitrile (%)	acetone (%)	methanol + 0.1% acetic acid (%)	hexane + 0.1 % acetic acid (%)
0	80		12	8
9.9	80		12	8
10	60	40		
17.9	60	40		
18	80		12	8
24	80		12	8

3. RESULTS AND DISCUSSION

3.1. Validation Procedure for Binary Systems. To validate the equipment and the experimental method, solubilities of the system squalene + sc-CO₂ were measured at 312.95 and 324.05 K and compared with literature data.^{31–35}

For the system mentioned before, solubility data previously published ($T = 313.00$ K) and the data obtained in this work at 312.95 K are shown in Figure 2. According to the results of the comparison, the solubility data obtained in this work are in good agreement with Martinez-Correa et al.,³¹ Catchpole et al.³³ and Brunner et al.³⁵ throughout the measurement range.

However, deviations between the data obtained in this work and those reported by Hernandez et al.³² and Al-Darmaki et al.³⁴ are observed. Moreover, comparing the data of these authors between them, large deviations are found, especially at high pressures. The solubility data obtained for squalene + sc-CO₂ mixture (in liquid–gas equilibrium) are shown in Table 4.

Table 4. Experimental Solubilities of Squalene (y) in sc-CO₂ (Liquid–Fluid Phase Equilibria) at Temperature (T) and Pressure (P)^a

T/K	P/MPa	squalene y (10^4)
312.95	13.50	13.544
	16.03	19.854
	19.42	27.907
	25.00	41.163
	32.00	55.163
324.05	10.16	1.325
	15.42	13.531

^aStandard uncertainties for temperature and pressure were evaluated to be $u(T) = 0.02$ K and $u(P) = 0.02$ MPa. Relative combined expanded uncertainty with a 0.95 level of confidence ($k = 2$) for the composition was estimated to be $U_r(y) = 0.051$.

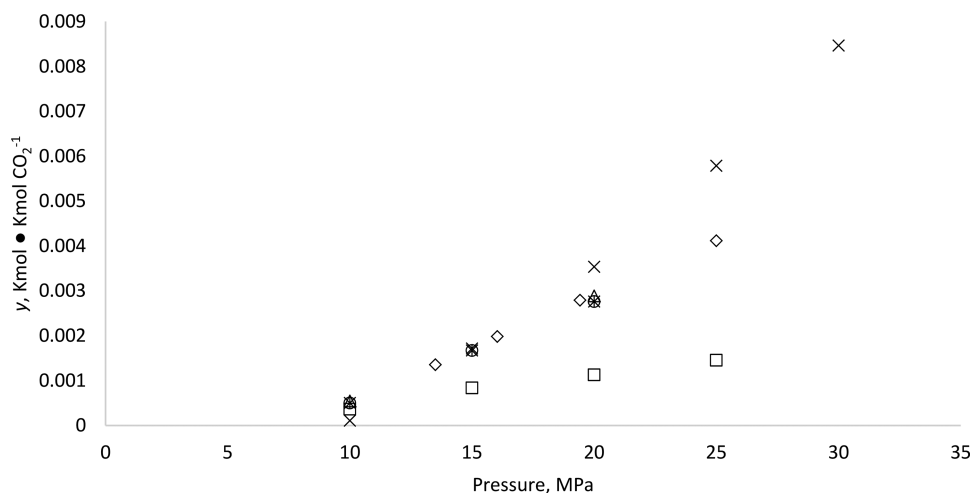


Figure 2. Solubility of squalene in sc-CO₂ at 313 K: (Δ) Martinez-Correa et al.;³¹ ($\times$) Hernandez et al.;³² ($*$) Catchpole et al.;³³ ($\square$) Al-Darmaki et al.;³⁴ ($\circ$) Brunner et al.;³⁵ ($\diamond$) this work.

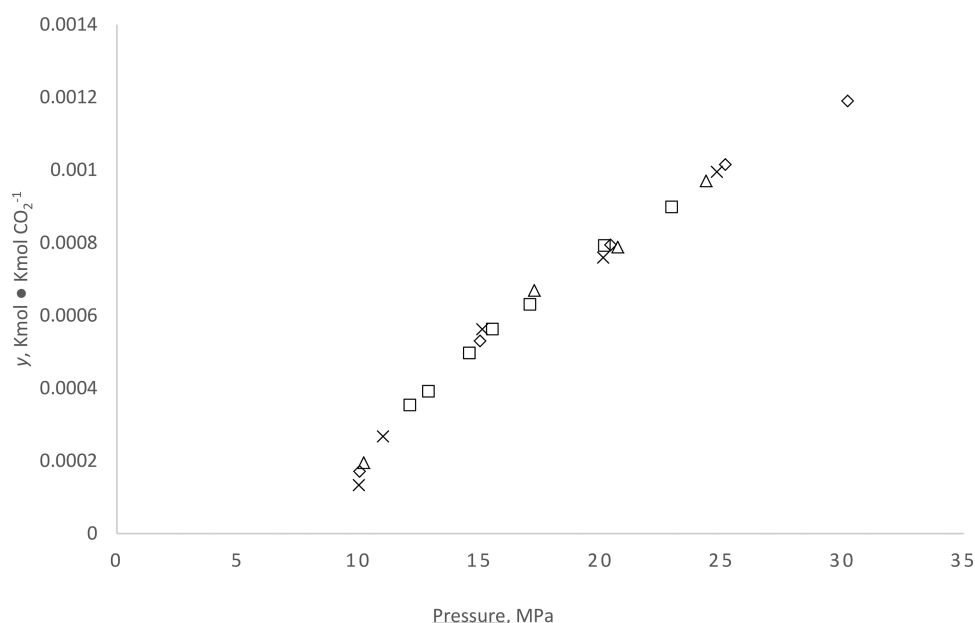


Figure 3. Solubility of Palmitic acid in sc-CO₂. (□) Brandt et al.³⁶ at 313 K; (×) Bamberger et al.³⁷ at 313 K; (△) Noh et al.³⁸ at 313.15 K; (◇) This work at 313.30 K.

Because of the deviations found in the squalene + sc-CO₂ mixture, a second binary system (palmitic acid + sc-CO₂) was measured at 313.30 K. The behavior of the solubility of the palmitic acid + sc-CO₂ obtained in this work, as well as different data of the international literature^{36–38} are shown in Figure 3. As it can be observed, the experimental data are in good agreement with the references consulted, which allows validation of the operation of the experimental equipment. Table 5 shows the experimental solubilities of palmitic acid in

Table 5. Experimental Solubilities of Palmitic Acid (y) in sc-CO₂ (Solid–Fluid Phase Equilibria) at Temperature (T) and Pressure (P)^a

T/K	P/MPa	palmitic acid y (10^4)
313.30	10.02	1.715
	15.01	5.305
	20.39	7.942
	25.14	10.153
	30.19	11.904

^aStandard uncertainties for temperature and pressure were evaluated to be $u(T) = 0.02$ K and $u(P) = 0.02$ MPa. Relative combined expanded uncertainty with a 0.95 level of confidence ($k = 2$) for the composition was estimated to be $U_r(y) = 0.051$.

sc-CO₂ (in solid–gas equilibrium). Once the procedures and the experimental method were validated, ternary systems containing squalene and fatty acids in sc-CO₂ were measured.

3.2. Comparison and New Data: Ternary Systems. Ternary systems containing squalene and oleic acid in sc-CO₂ were measured at 313.31, 323.29, and 333.18 K within a pressure range of 11.05–29.99 MPa.

The squalene + oleic acid + sc-CO₂ ternary system was previously reported in international literature⁴⁵ and measured in this work. Reference data compared correspond to a similar molar fraction of squalene in feed (0.4138); the original data can be seen in ref 45. Minimum deviations are found between the reference and the experimental data reported in this work; therefore, the trend of both sets of data are very similar. It is

worth mentioning that there is a difference (0.02 K) between the experimental temperature of this work and the reference temperature. Figure 4 shows the comparison between the solubility data for squalene + oleic acid + sc-CO₂ system from the international literature⁴⁵ with the experimental solubilities obtained in this work.

Squalene and oleic acid both are miscible liquids. Table 6 shows experimental results for squalene and palmitic acid in sc-CO₂ (in liquid–gas equilibrium). The mole fraction solubility range for squalene and oleic acid are $(0.01 \times 10^{-4}$ to 23.86×10^{-4}) and $(0.001 \times 10^{-4}$ to $25.77 \times 10^{-4})$, respectively. From the data given in Table 6 it can be observed that for all temperatures an increase in pressure causes an increment in solubility of squalene and oleic acid in sc-CO₂. This can be explained as being due to the increment on sc-CO₂ density by effect of an increment in pressure at fixed temperature. On the other hand, at 313.31 and 323.29 K and low pressures, the solubility of oleic acid is lower than squalene, but at high pressures, the solubility of oleic acid becomes greater than squalene. This phenomenon disappears in the isotherm of 333.18 K, where the solubility of squalene is always greater than oleic acid.

Finally, a new ternary system squalene (1) + palmitic acid (2) in sc-CO₂, was measured at 314.47, 324.28, and 333.39 K with initial mole fraction of $x_1 = 0.4208$. New data are shown in Table 7 and correspond to experimental solubility of squalene and oleic acid in sc-CO₂ (in solid–liquid–gas equilibrium). The solubility range for squalene and palmitic acid are $(0.01 \times 10^{-4}$ to $47.78 \times 10^{-4})$ and $(0.006 \times 10^{-4}$ to $11.59 \times 10^{-4})$ mole fraction, respectively. Figure 5 shows the data of binary systems at 312.95 and 313.30 K, and ternary systems at 314.47 K. For the ternary system squalene + palmitic acid in sc-CO₂, the same phenomenon is presented as the previous system, at 333.48 K a decrease, with respect to other temperatures, in the solubility of both compounds (squalene and palmitic acid) in the ternary system is obtained. However, the solubility of palmitic acid is always lower than that of squalene.

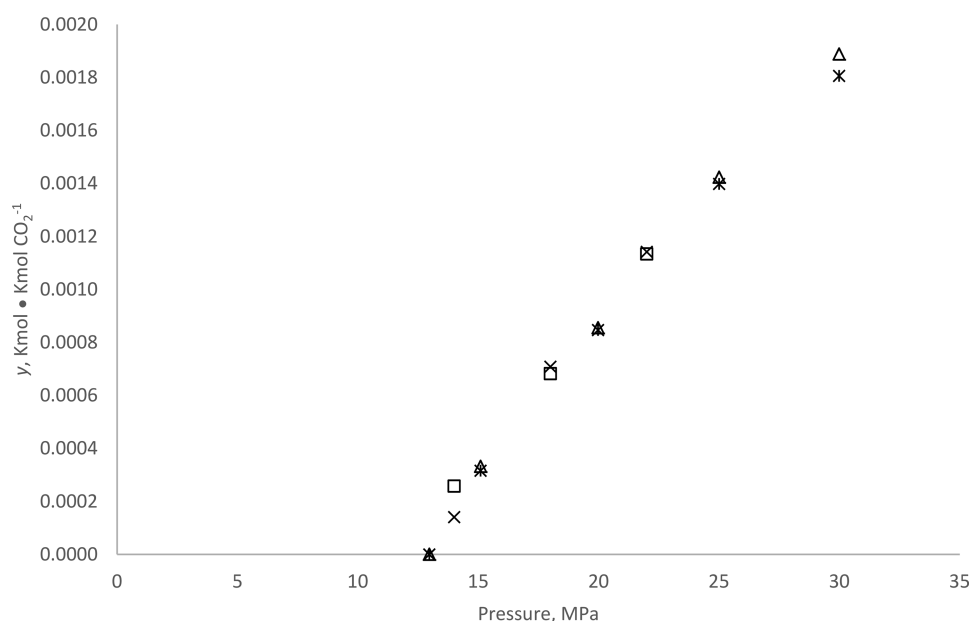


Figure 4. Solubility of squalene + oleic acid in sc-CO₂. Ruivo et al.⁴⁵ at 333.20 K: (□) squalene; (×) oleic acid. This work at 333.18 K: (△) squalene; (*) oleic acid.

Table 6. Experimental Solubilities of Squalene (y) and Oleic Acid (y_2) in sc-CO₂ (Liquid–Fluid Phase Equilibria), and Reference Densities of CO₂ (ρ_{ref}) at Temperature (T) and Pressure (P)^a

T / K	P / MPa	squalene y (10^4)	oleic acid y_2 (10^4)	ρ_{ref} /kg·m ⁻³
313.31	11.05	2.22	1.26	683.38
	14.83	8.05	6.79	776.34
	20.00	14.33	13.67	838.96
	25.02	19.56	20.53	878.92
	29.98	23.86	25.77	909.16
323.29	12.85	1.67	0.65	627.85
	14.98	4.99	4.12	698.02
	20.09	11.04	11.44	784.58
	24.98	16.14	17.88	833.37
	29.92	20.43	23.11	869.35
333.18	12.97	0.01	0.001	503.05
	15.10	3.33	3.16	607.49
	19.98	8.56	8.47	723.17
	25.01	14.24	13.98	786.51
	29.99	18.89	18.06	829.52

^aInitial mole fraction of squalene in the liquid phase (squalene (1) + oleic Acid (2)), $x_1 = 0.4138$. Standard uncertainties for temperature, pressure, and composition in the liquid phase were evaluated to be $u(T) = 0.02$ K, $u(P) = 0.02$ MPa and $u(x) = 0.0004$. Relative combined expanded uncertainty with a 0.95 level of confidence ($k = 2$) for the composition in the fluid phase was estimated to be $U_r(y) = 0.051$.

3.3. Modeling Experimental Data. Semiempirical density-based models are very useful to correlate experimental solubility data. In this work, all experimental data obtained (binary and ternary systems) were modeled by four density-based semiempirical models proposed by Chrastil,³⁹ del-Valle and Aguilera,⁴⁰ Bartle et al.,⁴¹ and Kumar and Johnston.⁴² For all models, the density of carbon dioxide was estimated using the equation proposed by Span and Wagner.⁴⁷

The absolute average relative deviation, AARD (%), was calculated by the following equation:

Table 7. Experimental Solubilities of Squalene (y) and Palmitic Acid (y_2) in sc-CO₂ (Solid–Liquid–Fluid Phase Equilibria), and Reference Densities of CO₂ (ρ_{ref}) at Temperature (T) and Pressure (P)^a

T / K	P / MPa	squalene y (10^4)	palmitic acid y_2 (10^4)	ρ_{ref} /kg·m ⁻³
314.47	9.98	2.73	1.00	597.12
	15.10	16.89	4.39	772.14
	20.25	29.01	7.29	835.17
	24.99	39.11	9.67	873.58
	29.99	47.78	11.59	904.70
324.28	10.02	1.18	0.56	369.66
	14.97	13.08	3.55	688.86
	20.00	24.58	6.23	777.70
	25.29	35.28	8.88	831.36
	30.01	43.18	10.68	865.95
333.39	10.00	0.01	0.006	288.60
	14.99	9.99	2.88	601.29
	20.18	19.29	4.99	725.03
	25.21	29.19	6.82	787.50
	30.01	36.87	8.01	828.80

^aInitial mole fraction of squalene in the mixture (squalene (1) + palmitic acid (2)) $x_1 = 0.4208$. Standard uncertainties for temperature, pressure and composition in the liquid phase were evaluated to be $u(T) = 0.02$ K, $u(P) = 0.02$ MPa and $u(x) = 0.0004$. Relative combined expanded uncertainty with a 0.95 level of confidence ($k = 2$) for the composition was estimated to be $U_r(y) = 0.051$.

$$\text{AARD} = \frac{100}{N} \sum_{i=1}^N \frac{|y_{\text{exp}} - y_{\text{cal}}|}{y_{\text{exp}}} \quad (1)$$

where N is the number of experimental data, y_{exp} is the experimental solubility, y_{cal} its corresponding solubility obtained from the correlation at the same condition. The results of experimental solubility data modeling for squalene, palmitic acid, and oleic acid in sc-CO₂ in binary and ternary systems are reported in Table 8, corresponding to each model calculated for the entire pressure and temperature range.

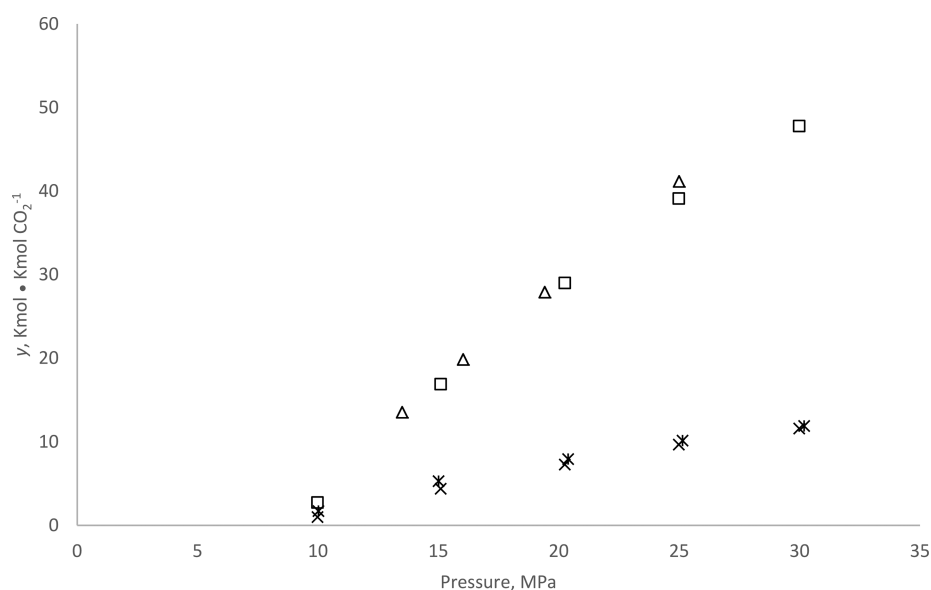


Figure 5. Experimental data for the systems: (Δ) squalene + sc- CO_2 at 312.95 K; (*) palmitic acid + sc- CO_2 at 313.30 K; ($\square$) squalene in ternary system (squalene + palmitic acid + sc- CO_2) at 314.47 K; ($\times$) palmitic acid in ternary system (squalene + palmitic acid + sc- CO_2) at 314.47 K.

Table 8. Results of Experimental Solubility Data Modeling for Squalene, Palmitic Acid, and Oleic Acid in sc- CO_2 in Binary and Ternary Systems^a

ref	model	system	solute	correlation constants				AARD%
				A	B	C	D	
Chrastil ³⁹	$\ln S = A + \frac{B}{T} + C \ln \rho$	SQ+ sc- CO_2	SQ	-22.081	-2852.040	5.131		3.23
		SQ+ PA + sc- CO_2	SQ	-20.383	-2491.600	2.650		9.22
		SQ+ PA + sc- CO_2	PA	-26.864	-688.369	2.518		10.54
		SQ+ OA + sc- CO_2	SQ	-8.716	-3746.720	3.380		1.07
		SQ+ OA + sc- CO_2	OA	-17.578	-2458.320	4.011		0.54
del-Valle and Aguilera ⁴⁰	$\ln S = A + B \ln \rho + \frac{C}{T} + \frac{D}{T^2}$	SQ+ sc- CO_2	SQ	-0.456	3.131	-2851.670	-60.893	3.23
		SQ+ PA + sc- CO_2	SQ	4.837	2.653	-2482.340	-1491.910	1.22
		SQ+ PA + sc- CO_2	PA	-677.027	2.529	4.343×10^7	-7.011×10^7	2.67
		SQ+ OA + sc- CO_2	SQ	-151.515	1.358	9.927×10^4	-1.665×10^7	1.41
		SQ+ OA + sc- CO_2	OA	-65.135	2.001	4.176×10^4	-7.147×10^6	0.69
Bartle et al., ⁴¹	$\ln \left(\frac{y_2^* P}{P_{\text{ref}}} \right) = A + \frac{B}{T} + C(\rho - \rho_{\text{ref}})$	SQ+ sc- CO_2	SQ	14.637	-5086.440	0.0098		2.74
		SQ+ PA + sc- CO_2	SQ	11.061	-4036.040	0.0088		1.88
		SQ+ PA + sc- CO_2	PA	4.174	-2233.550	0.0085		2.56
		SQ+ OA + sc- CO_2	SQ	-1.845	2.510	0.0055		1.11
		SQ+ OA + sc- CO_2	OA	13.486	-5074.320	0.0082		0.96
Kumar and Johnston ⁴²	$\ln(y_2) = A + B\rho + \frac{C}{T}$	SQ+ sc- CO_2	SQ	-1.7183	0.0075	-3173.240		1.26
		SQ+ PA + sc- CO_2	SQ	-2.7284	0.0066	-2748.090		1.05
		SQ+ PA + sc- CO_2	PA	-9.4535	0.0064	-945.601		3.20
		SQ+ OA + sc- CO_2	SQ	2.4333	0.0044	-4061.130		0.11
		SQ+ OA + sc- CO_2	OA	-2.0147	0.0058	-3033.860		0.09

^aNotation: SQ, squalene; PA, palmitic acid; OA, oleic acid; sc- CO_2 , supercritical carbon dioxide.

The first density-based model tested corresponds to Chrastil,³⁹ this model can be used for solids and liquids, based on chemical association law, and it is one of the most popular semiempirical models,⁴⁶ is a linear relationship between the logarithms of solubility and density. The best description can be found in the original paper. Figure 6A shows the experimental data and the data obtained by the model. This model has a greater deviation with the system squalene + palmitic acid + sc- CO_2 , and has less deviation with the system squalene + oleic acid + sc- CO_2 .

The del-Valle and Aguilera model is a modification performed to the Chrastil model with the addition of a second

order temperature term for a wider temperature range and solute solubility (less 100 g/L).⁴⁶ This model adequately represents the ternary systems, and has the greater deviation with the squalene + sc- CO_2 system (Figure 6B).

The model proposed by Bartle et al.,⁴¹ is in good agreement with experimental results for the ternary system (squalene + oleic acid + sc- CO_2), the deviation of this model increases for the binary system squalene + sc- CO_2 (Figure 6C).

Finally, the last model used was Kumar and Johnston,⁴² originally used to model solid solubility, but subsequently has been used to model experimental data from grape seed oil

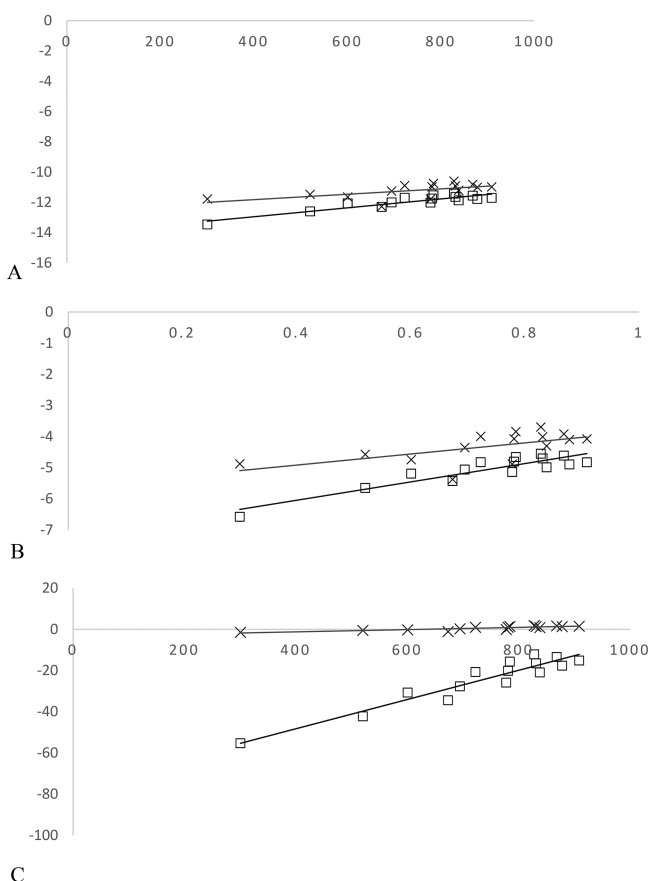


Figure 6. Comparison of the correlated results for squalene + palmitic acid + sc-CO₂ mixture: (□) squalene; (×) palmitic acid. (A) Chrastil model; (B) del-Valle and Aguilera model; (C) Bartle et al. model.

obtained by SCFE. This model has the lowest deviation with respect to all the models for the modeling of ternary systems.

4. CONCLUSIONS

The apparatus used in solubility of solids can be used in nonvolatile liquids as squalene and fatty acids below 30 MPa. The solubility data obtained in this work is in good agreement with international literature. Squalene solubility is greater than fatty acids in sc-CO₂. As expected, the solubility of ternary systems decrease in relation to binary system (squalene + palmitic acid + sc-CO₂).

Experimental data were correlated with semiempirical models, and the minimum AARD obtained for the binary system squalene + sc-CO₂ was 1.26% corresponding to Kumar and Johnston model; for squalene + palmitic acid + sc-CO₂ it was 2.67% from the del-Valle and Aguilera model; and for squalene + oleic acid + sc-CO₂ it was 0.11% from Chrastil model. These results can be used to develop an efficient process to extract squalene by SCFE.

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